Plantation Silviculture

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Plantation Silviculture (A10464)

Short Title:	Plantation Silviculture
Faculty	Science and Computing
Department	Science
Cluster	Forestry
Author	NMCCARTHY
Credits	5

Level: Intermediate

Description of Module / Aims

This module will introduce the whole concept of plantation forestry and the rationales behind it. Students will learn about production from plantation forests, and the economic, environmental, social and policy issues associated with practice. They will learn how to optimise the layout and design of plantations adhering to environmental constraints, and study specialised forms of plantation forest along with silvicultural systems.

Programmes

FORS-0013	BSc in Forestry (WD_SFORS_D)	3 / 6 / M

Indicative Content

- Introduction: the role of plantations
- Production and long-term productivity in plantations
- Rationales for plantation forestry: economic, biodiversity, and social and policy issues
- Layout and design of plantation forests
- Specialised forms of plantation: short rotation crops, plantations on disturbed land
- Silvicultural systems

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Interpret the concepts and reasons for plantation forests.
2. Debate the constraints on silviculture due to economic, ecological, and social factors.
3. Compare the layout and design of plantation forests.
4. Analyse the different types of silvicultural system.
5. Contrast specialised forms of plantations.
6. Analyse the management of a forest plantation with respect to its composition structure, biodiversity and protection.
7. Analyse relevant papers in the field of forest protection.

Learning and Teaching Methods

- Lectures.
- Field trips.
- Independent learning.
- Projects.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	60%	1,2,3,4,5
Continuous Assessment	40%	
Practical	30%	6
Case Studies	10%	7

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter. Lacking in technical ability/skill and understanding to carry out assessments. Failure to meet the objectives of assessments. Unable to carry out or submit independent study and research. Unable to present written work with attention to correct formatting, grammar and spelling.

40%-49%: Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on assessments. Attempt made to meet objectives and carry out independent study and research. Final result lacks balance and arguments undeveloped. Can present written work correctly formatted with minor grammar and spelling errors only.

50%-59%: Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out assessments. Satisfactory handling and awareness in carrying out independent study and research. Reporting and projects adequately structured and referenced.

60%-69%: Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design and current knowledge limitations when carrying out assessments. Projects and reporting coherent and soundly structured illustrating wide and in depth reading on subject coupled with the proper use of references.

70%-100%: Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject. Demonstrate imaginative approach, excellent technical ability, awareness, and aptitude in carrying out and reporting on assessments. Show initiative, individual thought, and enthusiasm in self study and independent learning.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	24
Field Trips	24	24
Independent Learning	87	87

Essential Material

- Hodge, S.J. _Creating and managing woodlands around towns_. UK: HMSO, 1995.
- Humphrey, S.J. _Grazing as a management tool_. UK: HMSO, 2000.

- Savill, P., J. Evans, D. Auclair and J. Falck. _Plantation Silviculture in Europe._. UK: Oxford University Press, 1997.

Supplementary Material

- Horgan, T., M. Keane, R. Mc Carthy, M. Lally and D. Thompson. _A guide to forest Tree species Selection and Silviculture in Ireland_. Ireland: COFORD, 2003.
- Matthews, J.D. _Silvicultural Systems_. UK: Oxford, 1997.
- Plachter, H. and U. Hampicke. _Large-scale__Livestock Grazing: A Management Tool for Nature Conservation_ . UK: Springer, 2014.

Requested Resources

- Lecture Room: Loose Seated